Lab 4 Report – Hilbert Transform

1. make a spectrogram of the original signal (NFFT = 212, Hamming window, 90% overlap)

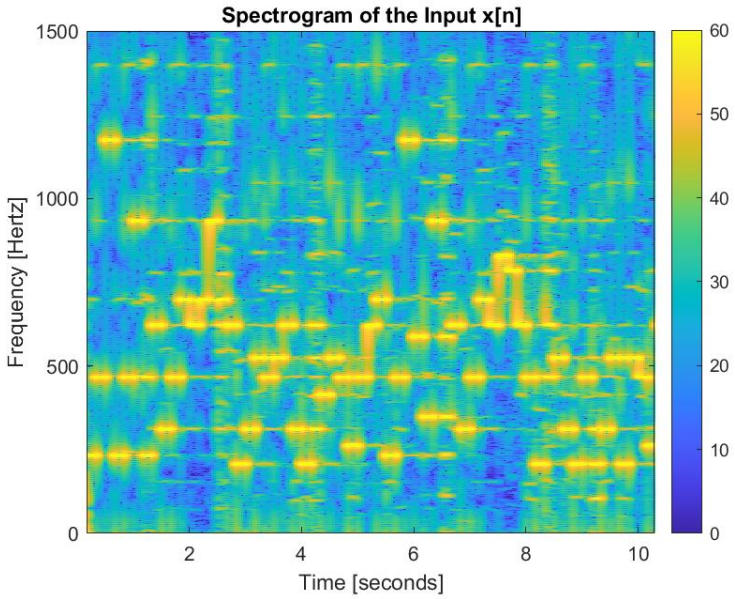


Figure 1: Spectrogram of the input audio

2. overlay the filtered instantaneous frequency over the top of the spectrogram.

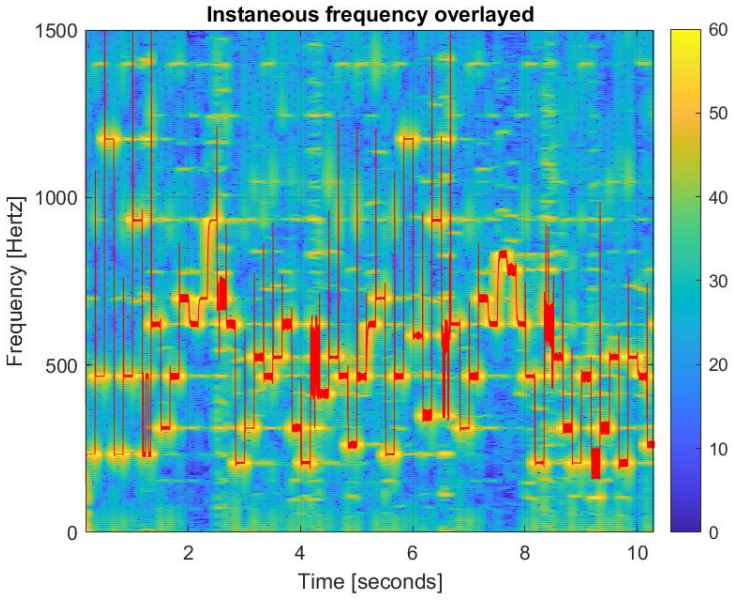


Figure 2: Instantaneous frequency over the spectrogram of the input.

3. describe the relationship between the instantaneous frequency and the spectrogram.

   The instantaneous frequency contains the exact information about which notes are being played. While the spectrogram contains harmonics of the input.

4. plot the magnitude and phase response for the Hilbert filter h(n) and the delay filter d(n)

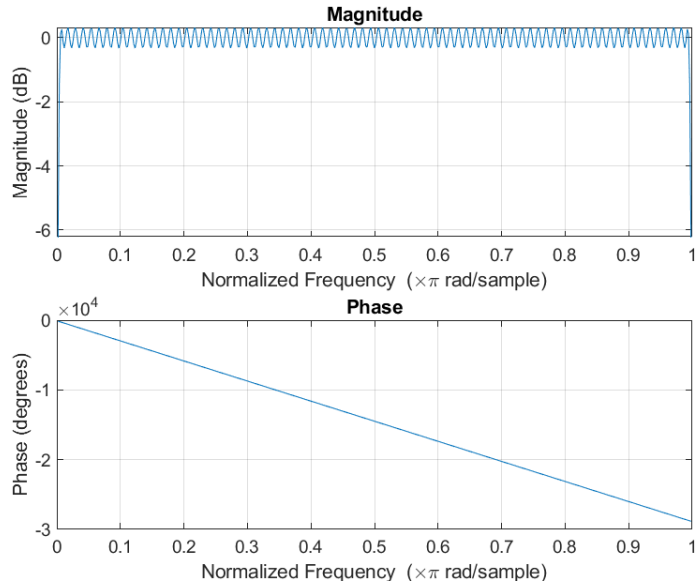


Figure 3: Magnitude and phase of the Hilbert filter

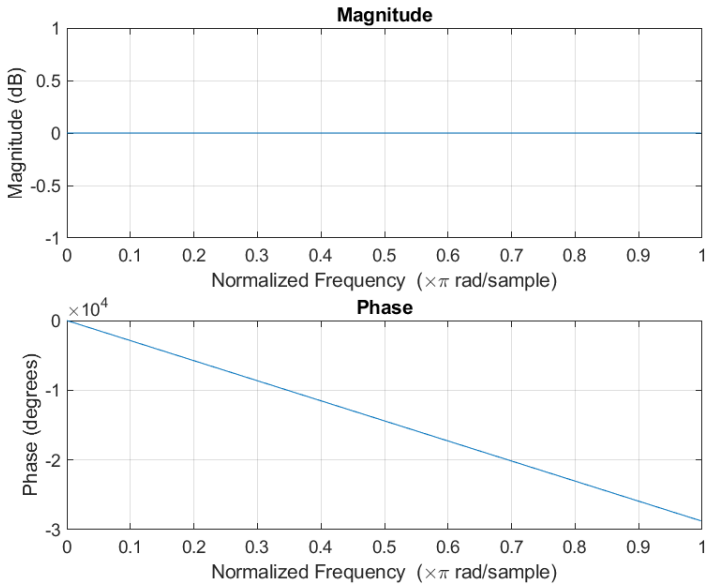


Figure 4: Magnitude and phase of the delay filter

5. plot the magnitude and phase of the Hilbert transformer d(n) + j*h(n)

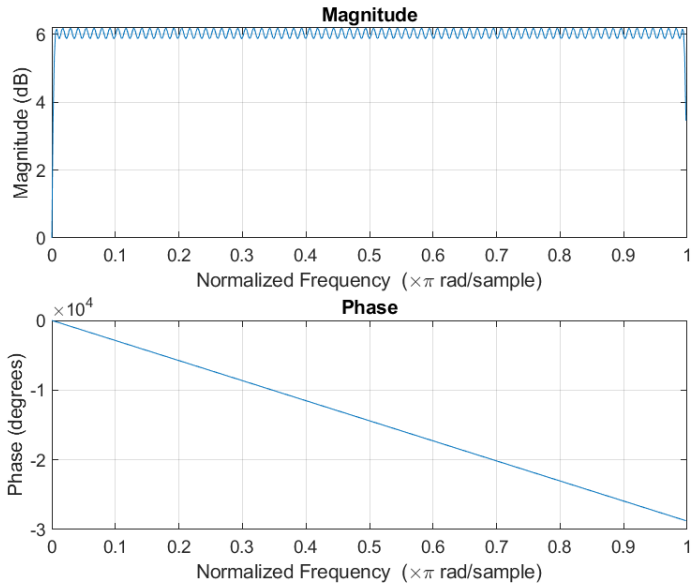


Figure 5: Magnitude and phase of the Hilbert transformer

6. describe in words the action of the Hilbert transformer based on the frequency response

   The Hilbert transformer is able to take a cos wave and turn it into a single frequency complex exponential.

7. plot the magnitude and phase of the cascade combination of the 1st-order difference filter and the Gaussian filter.

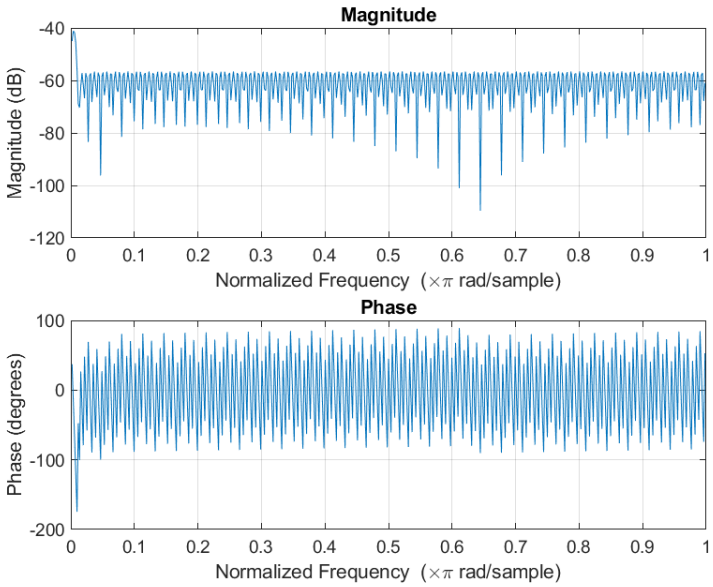


Figure 6: Magnitude and phase of the cascaded gaussian filter and difference filter.